

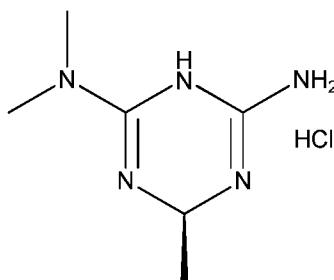
**“PROCESS FOR PREPARATION OF IMEGLIMIN AND SALTS
THEREOF”**

FILED OF THE INVENTION:

The present invention relates to an improved process for the preparation of Imeglimin & salts thereof. More particularly, the present invention relates to chiral resolution of Racemic Imeglimin and its salts which is industrially advantageous and economically significant.

BACKGROUND OF THE INVENTION:

Imeglimin is chemically known as (6R)-(+)-4-dimethylamino-2-imino-6-methyl-1,2,5,6-tetrahydro-1,3,5-triazine hydrochloride, having the structure compound of formula (I).



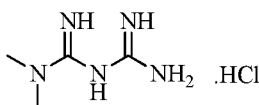
Formula-I

Imeglimin was first discovered by Poxel and developed in partnership with Sumitomo Dainippon Pharma Co. Ltd, launched with the brand name of Twymeeg® in Japan for the treatment of type-2 diabetes mellitus. Imeglimin was first approved in Japan on June 23rd, 2021 for the treatment of type-2 diabetes mellitus. Imeglimin is under phase-3 clinical trial in USA & Europe.

US7034021B2 first discloses the Imeglimin. Further US7034021B2 does not disclose process for preparation of Imeglimin by any specific example. US7501511B2 (herein after US'511) discloses resolution process for preparation

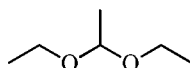
In line with the above objective, the present invention provides a process for preparation of Imeglimin HCl compound of formula (I), which process comprises;

a) reacting compound of formula (II)



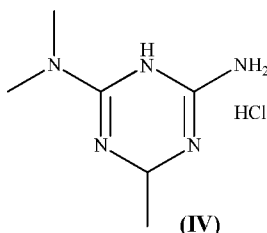
(II)

with a compound of formula (III) in a solvent



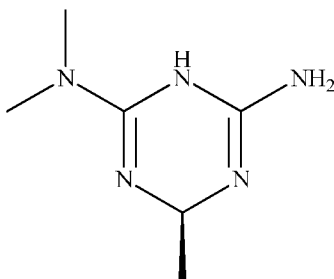
(III)

in presence of p-toluene sulphonic acid monohydrate under suitable condition to obtain racemic compound of formula (IV),



(IV)

b) reacting the racemic compound of formula (IV) with a suitable chiral amino acid or its derivative in the presence of a base and a solvent to obtain R-Imeglimin L-amino acid salt compound of formula (V),



R-Imeglimin -L- amino acid salt

(V)